The broadband call control demonstrator — a platform for ITU-T, DAVIC and TINA-C implementations

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The paper firstly provides an introduction to the broadband call control demonstrator platform developed at BT Laboratories. It describes the basic connection level functionality used to give point-to-point and point-to-multipoint call demonstrations. It then describes the introduction of a Digital Audio-Visual Council (DAVIC)-conformant session control capability and how this is used as the basis of a more comprehensive demonstration. The paper also examines how future networking concepts such as object orientation and distributed processing environments are being introduced, in particular the interworking between DAVIC and Telecommunications Information Network Architecture Consortium (TINA-C) approaches. Finally, some of the evolutionary aspects of the broadband call control demonstrator platform are introduced.

1. Introduction

In order to provide future broadband services in a timely and efficient manner, a network is needed which is both powerful and flexible. Key to this requirement is the need to form broadband connections on-demand, with specific bandwidth and quality of service characteristics. To be in a position to better understand this level of 'control', work has been under way investigating the requirements of future networks and their associated signalling systems. A key part of this work has involved the development of a broadband call control demonstrator platform, on which new concepts could be investigated. The platform has proved to be a powerful tool in terms of understanding the operation and interworking of signalling protocols. Its origin, current state and future evolution are covered in this paper.

Specifically, the approach taken to understand the requirements of 'broadband call control' is presented, covering the evolution of the broadband call control demonstrator, starting with the low-level call/connection control mechanisms and then examining the higher layer session control. Ultimately future object-oriented and object request broker techniques are looked at.

The original purpose of the demonstrator platform was as a learning tool to develop expertise, but has since evolved into a comprehensive foundation for other teams to explore broadband networking concepts. Its main focus is still to concentrate on call control issues and how they would be adopted in a full-scale network deployment.

1.1 Signalling architecture adopted by the broadband call control demonstrator

Later sections describe in greater detail the aspects of signalling and control explored within the project. This section introduces the basic concepts and architecture of the broadband call control demonstrator.

In order to simplify the approach to signalling in a large complex network, the end-to-end connectivity can be broken down into simpler elements.

At the lowest level are the elementary point-to-point, link-by-link 'connections'.

Slightly higher, an end-to-end view is given by concatenating the links to create a 'call'. Currently the call/connection levels are considered as a single layer, controlled using a single signalling protocol as described later. This supports a single point-to-point connection whereby either party can initiate or clear the connection.

Using the architecture developed within the Digital Audio-Visual Council (DAVIC) [1], the layers of control can be represented as shown in Fig 1. The call/connection layer provided the 'S4' control layer of the demonstrator.

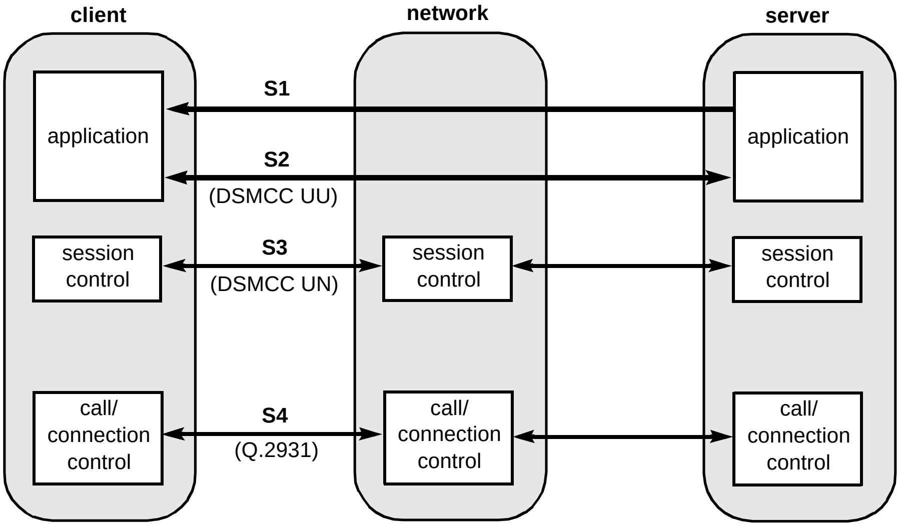

Fig 1 Broadband call control demonstrator — control architecture.

Moving higher still, a set of associated calls can be aggregated into a 'session'. This view allows more complex services to be co-ordinated, where connections may be set up, released or transferred, etc, during an active session. To provide this role the team adopted the use of DAVIC session control. This uses the digital storage media command and control set (DSMCC) user-to-network (U-N) protocol [2], and allows an abstraction of the complex call/ connection control mechanisms.

The DSMCC U-N protocol provided the 'S3' session control layer of the demonstrator (see Fig 1).

End-to-end user information is exchanged over the 'S1' and 'S2' information flows. The 'S2' flow, provided by DSMCC U-U (user-to-user) protocol, supports the bi-directional user-to-user control, used for browsing and controlling service streams. The 'S1' information flow is the service stream itself, e.g. a Moving Picture Experts Group (MPEG) video stream.

1.2 Basic demonstrator platform

The requirement was to develop expertise in the emerging signalling protocols being supported within the International Telecommunications Union (ITU-T) and the ATM Forum. At the time work started on the broadband call control demonstrator, the asynchronous transfer mode (ATM) switches available only supported permanent virtual channel (PVC) functionality. In order to support switched virtual channels (SVCs), there were effectively three options:

- to use a proprietary signalling mechanism, such as SPANS (simple protocol for ATM networks),
- to buy third-party software which could be ported on to the relevant switch platform,
- to write the software to implement a 'standards-based' signalling protocol.

It was decided to implement a sub-set of the ITU-T Q.2931 standard, which was developed on a number of Sun workstations to represent both the user and the network side functionality. Figure 2 represents the basic architecture adopted.

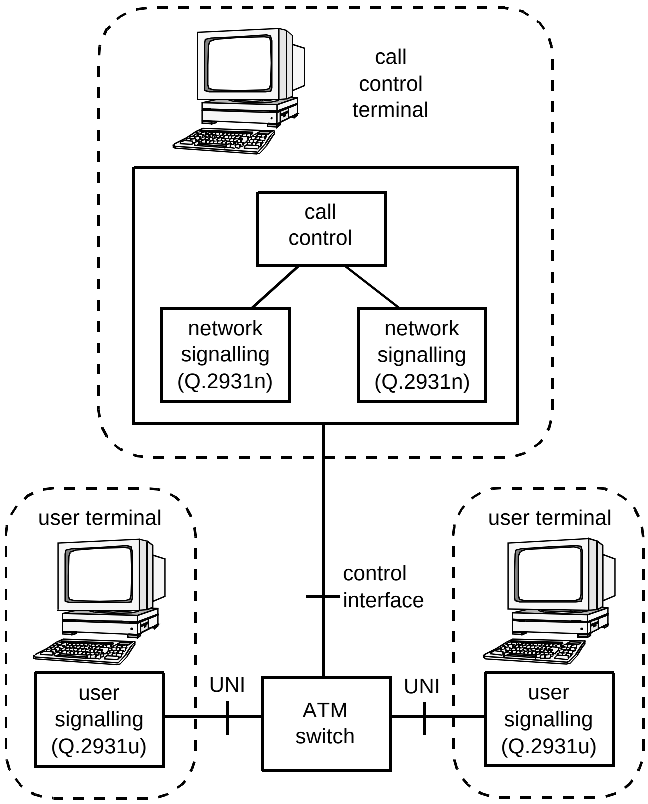

Fig 2 Physical demonstrator architecture.

2. Switch control

The control software developed by the team implemented signalling, call control, address server, connection acceptance control (CAC), connection control and resource manager functionality. The ATM switch was controlled by a simple network management protocol (SNMP)-based interface that allowed connections to be configured across the ATM switch at the request of the switch control software.

2.1 Switch control architecture

Using the switch control architecture as described in the RACE 1044 TRIBUNE [3] (which provided one of the first implementations of user/network interface (UNI) signalling) as the basis, a switch control architecture was developed.

The architecture (see Fig 3) was designed in a modular fashion with defined interfaces between the functions. Once completed, the architecture allowed for the flexible exploration of broadband call control issues independent of the ATM switch software and flexible enhancement to support additional functionality.

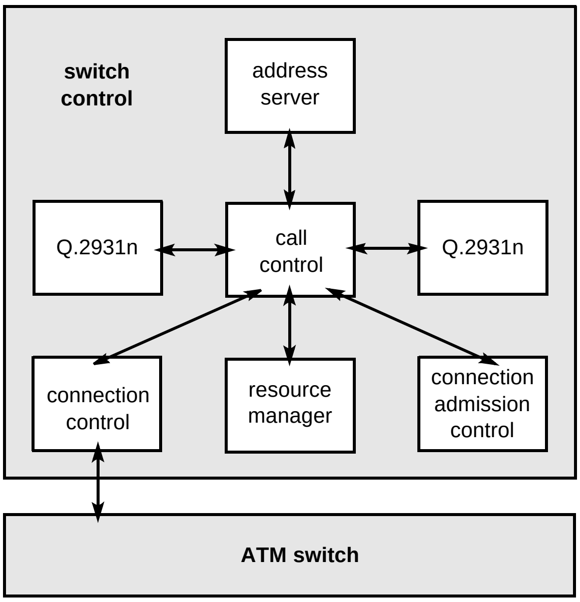

Fig 3 Switch control architecture.

There are three phases in the configuration of an ATM connection — reservation, allocation and release. Figure 4 shows the interaction between the switch control functions during the reservation phase.

The functions are described in more detail in the sections below.

2.2 Call control

Call control co-ordinates the reservation, allocation and release of resources. On receipt of an ATM signalling message the appropriate internal messages are generated to reserve, allocate or release the requested resources. Once complete the appropriate ATM signalling message is generated.

2.3 Address server

The ATM signalling messages contain a calling party number and called party number that must be mapped to physical ports and VPs on the ATM switch. The address server contains a database which is manually configured with these numbers and their associated physical ports and virtual paths (VPs).

2.4 Connection acceptance control (CAC)

The connection acceptance control uses the quality of service (QoS) parameters from the ATM signalling messages to calculate an effective bandwidth across the ATM switch. This effective bandwidth is then used in requests to the resource manager to see if there is enough free resource to accept the connection. To demonstrate the basic operation of the CAC process two algorithms were implemented — peak cell rate and linear. The peak cell rate algorithm only makes use of the peak cell rate parameter in the ATM signalling message. The linear algorithm makes use of the peak and sustainable cell rate to calculate an effective bandwidth.

2.5 Resource manager

The resource manager keeps track of ATM switch resources in terms of bandwidth, virtual paths (VPs) and virtual channels (VCs). Requests for resource are mapped against what is currently allocated and a decision given on whether the request can be accepted. The resource manager also provides an appropriate VPI and free VCI for both incoming and outgoing links.

2.6 Connection manager

The connection manager is responsible for mapping the requirement of the switch control software in terms of ports, VPs, VCs and bandwidth into a request to the ATM switch for a connection using those parameters. If support for different vendors' ATM switches were required, the flexibility of the switch control software means that this would be the only interface requiring modification.

3. Call/connection control

This section outlines the key features implemented on the broadband call control demonstrator, areas that caused problems, and where lessons were learnt. A more detailed description of the operation of broadband signalling protocols is covered by Law [4] and Jones [5].

3.1 Point-to-point connection control

The initial demonstrator phase was only required to provide a simple point-to-point switched virtual channel connection. This could be supported by Q.2931 [3].

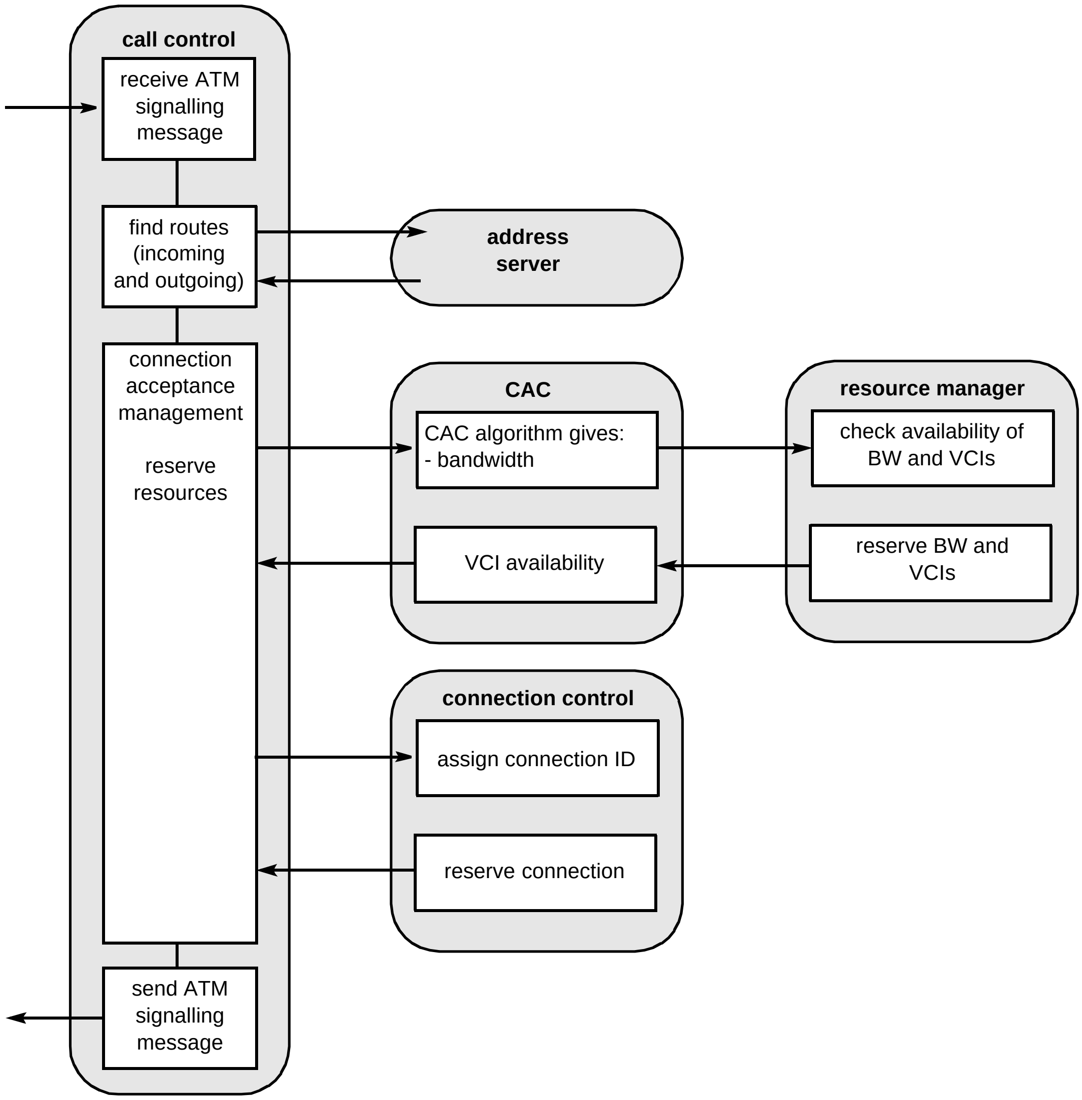

Fig 4 Example message flows — reservation.

It was decided from the outset of the project that little would be achieved by implementing every feature of the protocol, especially those relating to time-out and release ‘causes’. A sub-set was developed, covering the features listed below:

- single-sided call establishment,
- both ends able to release connections,
- simple bearer channel established across an ATM switch,
- network release of the call request in the event of incorrect call set-up.

Figure 5 illustrates the basic call flows to establish a point-to-point connection. It represents the information exchange between the user-side signalling entity (Q.2931u) and the network-side signalling entity (Q.2931n).

The figure illustrates the particular implementation of the CALL PROCEEDING message which is shown to originate at the called terminal. Generally this message would be originated by the network.

3.2 *Point-to-multipoint connections*

The next phase of development was the enhancement to support point-to-multipoint, unidirectional connections by implementing the additional functionality defined in ITU-T Q.2971.

This configuration was used to model a ‘video distribution’ application, where a ‘root’ terminal was able to add ‘leaf’ terminals to an existing connection. This is represented by the information flows in Fig 6.

3.3 *Leaf-initiated join (LIJ) capability*

A natural extension of the point-to-multipoint connection was the implementation of the ATM Forum leaf-initiated join procedure, which allowed the leaves themselves to request connection to an existing call.

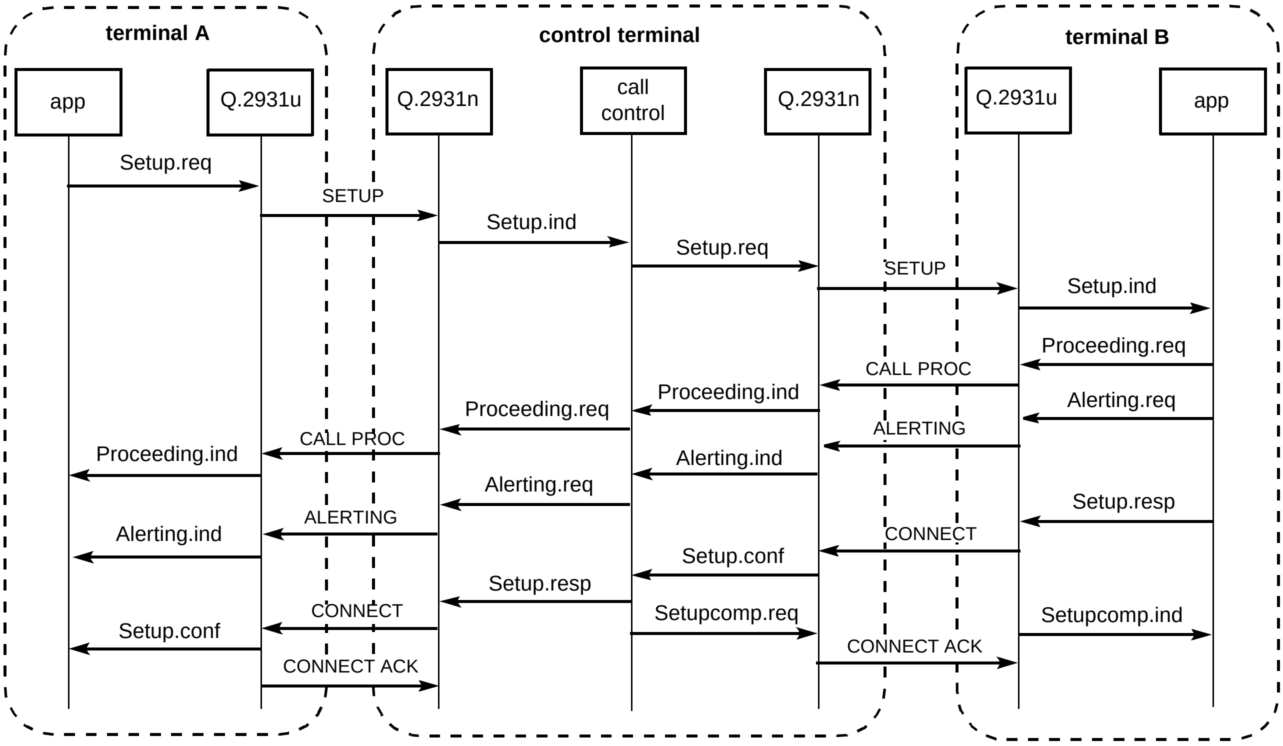

Fig 5 Message flows for point-to-point connection.

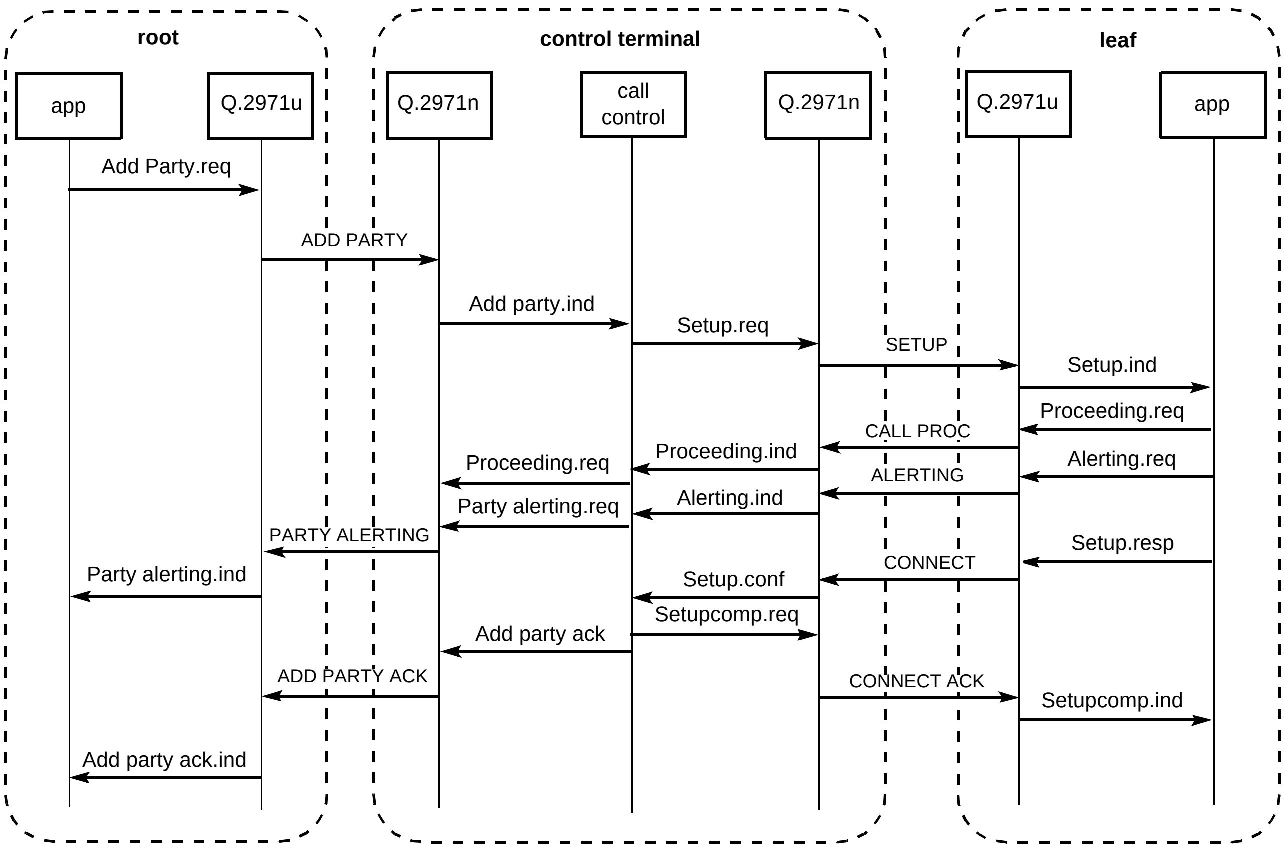

Fig 6 Point-to-multipoint information flow — adding a party.

This appears a simple addition, though certain problems emerge when this is migrated to a larger scale network:

- the leaf terminals need to identify which particular call they wish to join — a set of default values could be provided for 'known' or frequently used connections, but those formed dynamically would be more difficult to identify,
- the point-to-multipoint and LIJ capabilities described so far only support unidirectional connections — when these later evolve to bi-directional connections there needs to be some form of control to prevent the root terminal becoming congested.

3.4 *The role of proxy signalling*

Proxy signalling (see Fig 7) is a mechanism which allows a network element to handle signalling on behalf of a terminal which does not possess full signalling capabilities. There are several approaches for this, the one represented by this stage of the demonstrator allowed a network element, e.g. a video server, to be controlled by the root terminal initiating the point-to-multipoint connections. This is represented in Fig 7 which shows a proxy UNI from the root terminal handling the signalling messages, while the data connections are made directly to the video server itself.

A more realistic scenario would be where a piece of a customer's equipment without full user signalling capabilities, e.g. a set-top box (STB), would use a centralised proxy signalling agent (PSA) to establish connections. This scenario is shown in Fig 8.

The user's terminal or STB would either use a lightweight signalling protocol, or adopt a higher level view and request the establishment of a 'session' as described below in section 4, and shown in Fig 9.

The use of proxy signalling highlights the requirement for information to be exchanged between the session (S3) and the call/connection (S4) layers. In particular, the session layer must be informed of the lower layer connections that have been established in order to instruct the end user which VPI/VCI values to use for the S1 and S2 application flows. This information is returned in a resource descriptor

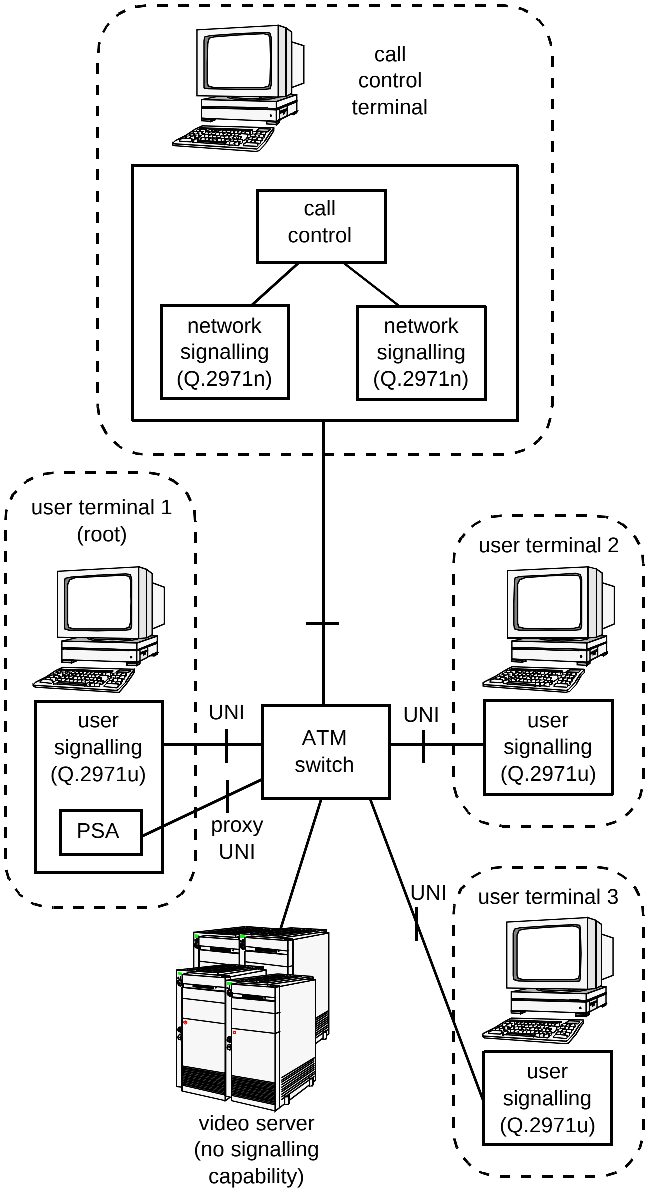

Fig 7 Proxy signalling on behalf of a network element (video server).

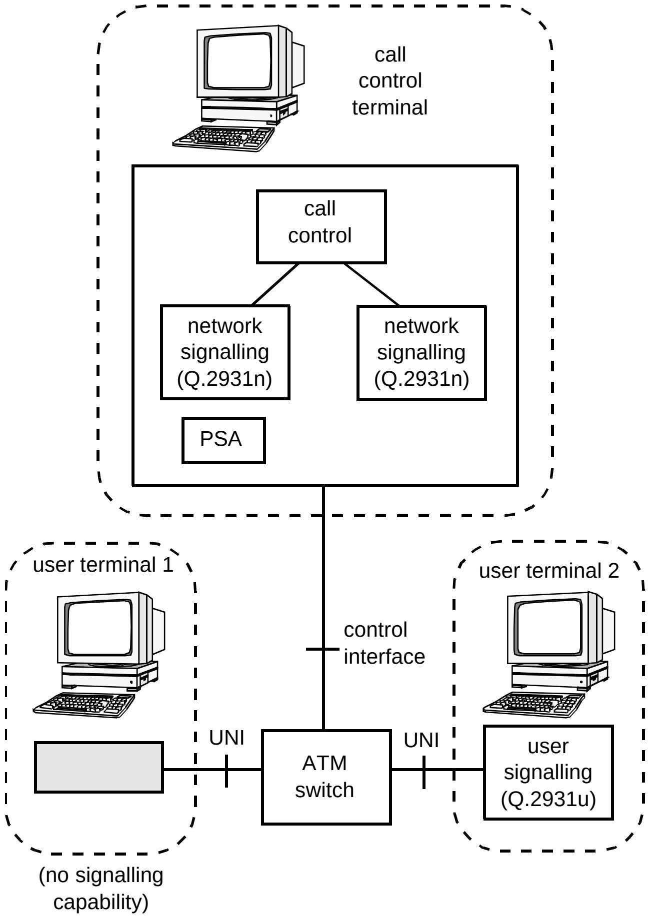

Fig 8 Proxy signalling on behalf of a user's terminal.

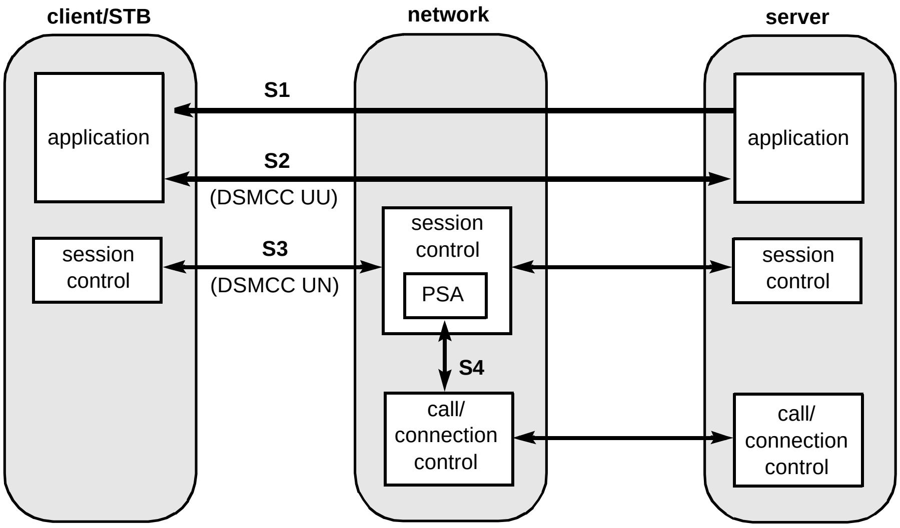

Fig 9 Proxy signalling performed by session controller.

information element as part of the 'ClientSessionSetup Confirm' message as shown in Fig 10.

Figure 10 identifies the following stages, labelled (a) to (d), where additional processing is required.:

(a) Upon initialisation, the user's terminal needs to inform the network that proxy signalling is required. This will enable the network to route the connection-layer signalling messages to the proxy agent and not back to the terminal itself.

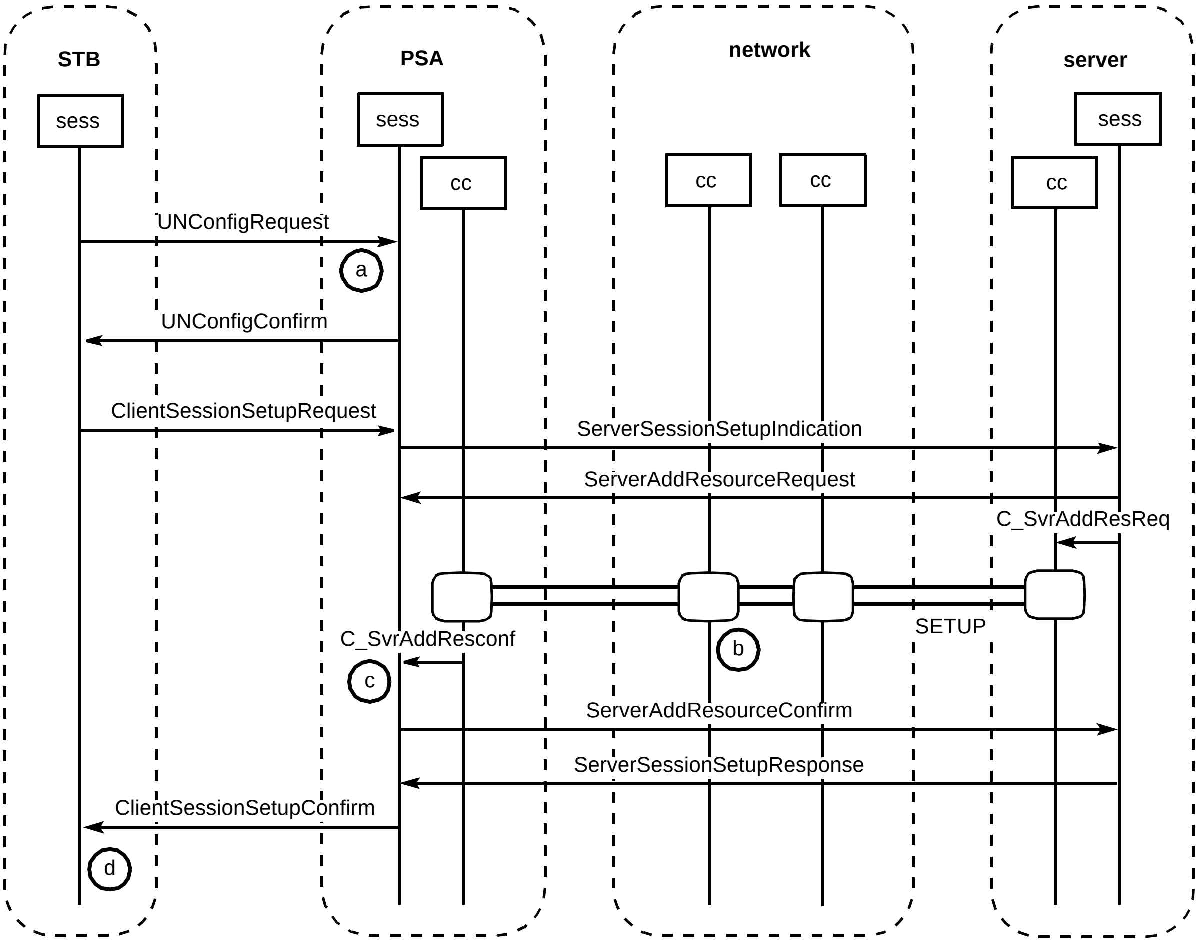

Fig 10 Information flows for PSA operation.

(b) The network needs to insert the VPI/VCI values used for the S1 and S2 connections into the ‘call/connection level’ message returned to the proxy agent.

(c) The proxy agent needs to extract the VPI/VCI values from the connection layer message and insert them into a resource descriptor information element within the ClientSessionSetupConfirm message.

(d) The user extracts the VPI/VCI values from the session layer message to identify which channels the S1 and S2 information will use.

3.5 Call/connection control application programing interface (API)

A simple connection control API was developed to allow higher (application) layers to easily access the connection capabilities offered by the demonstrator. It comprised a set of primitives, based on the Q.2931 message format, which allowed the application to request specific parameters for its connections.

4. Session control

After the call/connection control and switch control functionality had been implemented the next ke y area to consider was the little understood area of session control. This provides continuity of interactions between users, service providers and network providers when the delivery of a particular service is implemented through more than one call/connection. From the network’s point of view, session control consists of a number of related call/ connections which occur consecutively to provide some service to a user. It should be noted that a particular application may use its own session controller to control a number of (application-) related calls (call control terminal entities or agents), but there is no requirement for the network to relate these concurrent calls. The session control described in the DAVIC specification was based on ISO DSM-CC [2].

4.1 ‘Methods’ of session control

There are several methods of using DSM-CC within an ATM network. The methods differ in the way that the session (as identified by the SessionId and Resource Number) is tied to the call/connection (as identified by the CallReference). The four methods are discussed below.

- Session method

  The server informs the network (session and resource manager (SRM)) of the required session using the ServerAddResourceRequest message, and the network initiates the connection. The Q.2931 messages initiated by the network include the resourceId (SessionId + ResourceNumber).

- Network method (with AddResource message)

  The server allocates resources in the network prior to sending an AddResource message to the SRM. The Q.2931 messages initiated by the server include the resourceId (SessionId + ResourceNumber).

- Network method (without AddResource message)

  This allows any server to initiate Q.2931 signalling to the network. The SETUP message sent by the server includes a resourceId (SessionId + ResourceNumber) and the ResourceGroupTag. The SRM is in the signalling path to allow it to perform resource management within a session.

- Integrated method

  Here the DSM-CC U-N messages are mapped into the Q.2931 signalling messages. DSM-CC U-N messages that are not naturally coupled with Q.2931 signalling messages can be handled in Q.2932 facility messages. The SRM is in the signalling path for the Q.2931 signalling messages and thus manages the session and resources.

DAVIC does not explicitly use one of the methods; rather its session control concept is based loosely on the network method (with AddResource messages). To ensure evolution of the session control protocol it was decided to follow the DSM-CC standard as opposed to the DAVIC proprietary implementation of DSM-CC U-N.

4.2 Roles

DAVIC identifies three key roles in its DAVIC V1.0 specifications — service consumer, delivery system and service provider. Through consideration of long-term business model work within TINA-C and ITU-T, it was felt that the DAVIC delivery system could in fact be separated into three roles — network connectivity provider, retailer and broker. Definitions for each of these roles are given below:

- service consumer — enrols for and uses services,
- network connectivity provider — provides the communications services that consist of calls and connections,
- retailer — provides the contact point for the consumer to arrange for, and the contact point for the service provider to offer, services which employ communications services as the delivery mechanism,
- broker — provides location information that the business domains use to locate other entities for the purpose of establishing communications,

- service provider — supplies a variety of services which employ communications as the delivery mechanism.

4.3 Information flows

The information flows given below (and shown in Fig 11) concentrate on the session control information flows. Other information flows are appropriate to the presentation, application and call/connection control levels.

Session set-up

The main information flows are those between the session layer entities (sess), carrying DSMCC U-N messages to establish the session. The primitives passed from the end-to-end (E-E) control layer down to the session layer do not form part of the DAVIC/DSMCC specification, but are proprietary, based on the DSMCC U-N messages they trigger, e.g. the E_initiate_ses primitive is based upon the ClientSessionSetupRequest message as defined within the DSMCC U-N specification.

Session release

The release phase is initiated by the service consumer system generating an E_initiate_rel primitive which is passed to the session layer control element, which in turn generates a ClientReleaseRequest message (see Fig 12).

Session transfer

The session transfer capability is supported via a session set-up and a session release.

4.4 Session control API

As had been implemented earlier with the call/connection control a high-level API to the session control level was defined, for use by the applications that would use this level. This API allows others to independently develop applications using the session control level.

5. The Telecommunications Information Network Architecture Consortium (TINA-C)

Following the work described above, the broadband call control demonstrator provided a complete implementation of a DAVIC-compliant system for supporting a video-on-demand (VoD) service using proxy signalling. The architecture to that point had evolved through a traditional standards-based approach, building on low-level

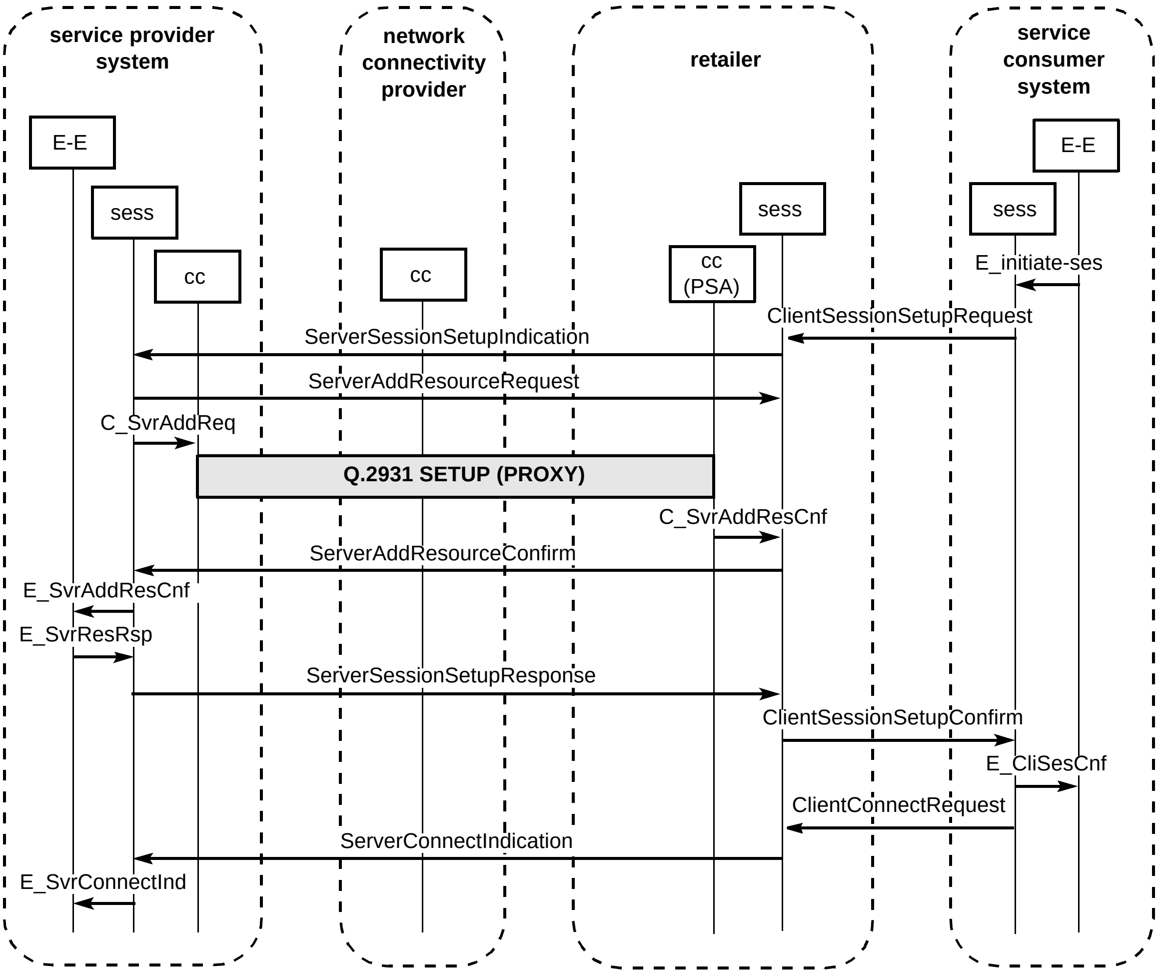

Fig 11 Session set-up.

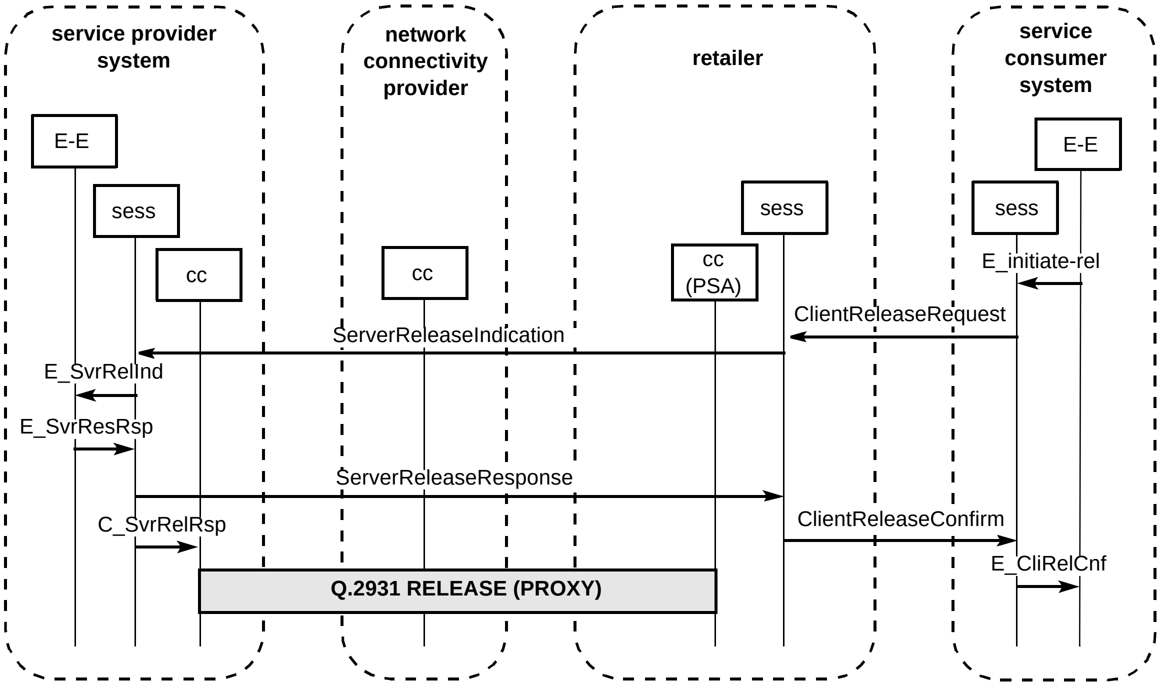

Fig 12 Session release.

connection capabilities to offer more complex services. It was decided to progress with a more revolutionary approach by considering the demonstrator in the light of work done within TINA-C.

5.1 *Introduction to TINA-C*

With the current speed of technical innovation, service operators have the challenge of incorporating new technologies within their networks and providing new services:

- with minimum development time,
- at minimum costs,
- in a flexible and manageable way.

In 1992 the world's leading network operators, telecommunications and computer manufacturers formed a consortium to develop a common software architecture for multimedia services to support these aims — the Telecommunications Information Network Architecture Consortium.

The initial work of TINA-C centred on the development of a software architecture, encompassing all parts of telecommunications and information systems. This architecture is based on the principles of object-oriented (OO) design, the decoupling of software components and the development of an 'enterprise model' showing how different stakeholders (companies) can interact to develop a service. TINA specifies three sub-architectures.

- Computing architecture

  This defines a distributed processing environment (DPE) in order to separate service applications from the computing platforms on which they run. With the DPE, computational objects can interact with each other without needing to know the physical location or computational platform of their peer.

- Service architecture

  This defines a set of principles for providing services and is based on the concept of a session which represents the information required for the provision of a service for a given duration, and the separation of objects into generic (common to all services) and service specific.

- Network architecture

  This defines a technology-independent model for establishing connections and managing telecommunications networks.

Latterly TINA-C has concentrated on the specification of a set of reference points between the different parties in a service, such as the consumer, connectivity provider, service provider, etc.

The architecture defined by TINA-C has been implemented in a number of validation projects in member companies, and TINA-C interacts with other standards bodies and industry consortia including ATM-F, DAVIC, ITU-T and Object Management Group (OMG) to provide harmony and avoid duplication of effort.

TINA-C has produced specifications on architectures and reference points, and work continues to develop and enhance these specifications. The TINA-based services have been implemented in trials by member companies, and some TINA-compliant products are on the market.

5.2 TINA and the broadband call control demonstrator

Given that TINA might move into a commercial phase, it becomes necessary to consider how one might interwork TINA-compliant products with legacy (non-TINA) systems. TINA customers may wish to access non-TINA services, TINA service suppliers may wish to support non-TINA customers, and network operators may have to link networks based on TINA and non-TINA architectures. As a means of exploring these issues it was decided to develop the broadband call control demonstrator to support one or more TINA components. The problems exposed in interworking DAVIC and TINA would be representative of generic interworking issues.

An initial problem faced by the development team was their lack of practical familiarity with TINA concepts and architectures. To address this issue, work was undertaken to implement one of the demonstrator components (the retailer) on a distributed computing environment. This would serve to:

- introduce the team members to DPE technology,
- allow the issue of supporting legacy (non-OO) code on a DPE,
- be a significant first step to developing expertise in TINA.

The DPE chosen for the work was ORBIX, a CORBA-compliant object request broker (ORB) supplied by the company IONA. The Q.2931 state machines supporting call control and the DSM-CC U-N state machines supporting the session control were reworked so that they communicated using interface definition language (IDL) interfaces. These identify the set of procedure calls that the processes can support and define them in a way that allows the procedures to interwork over the DPE. A software interworking unit (IWU) was developed which supported sockets and an interface to an ATM card to enable the retailer to communicate to the rest of the network and in addition supported an IDL interface. This provided the connection between the legacy network and the DPE on which the modified software was supported.

Finally the single workstation of the retailer was replaced with a workstation and a PC in order to explore the issues of running the retail functionality over a number of dissimilar hardware platforms. The completed TINA retailer is shown in Fig 13.

The PC has a CORBA daemon and runs a single process, the user agent, which sends information to the monitor showing the state of messages being processed in the retailer. The Unix workstation runs the interworking unit to link the CORBA DPE to the non-CORBA environment, a GUI to allow the operator to view and control the retailer and various functions (dummy user agent, address registration) which are required by the legacy (non-CORBA) DSM-CC and Q.2931 state machines to run in the CORBA environment. Finally a debugging function was developed to allow testing of the retailer.

This modified retailer successfully interworked with the call control demonstrator and provided a test bed for the exploration of CORBA-related issues. However, it did not at this stage support any new functionality.

6. Future work

This section introduces a range of new topics that are under investigation using the broadband call control demonstrator. They form the basis of an ongoing development of expertise within the team.

6.1 IP resource reservation protocol

Work to date within the call control team, and specifically on the broadband call control demonstrator, has concentrated on a ‘native ATM’ approach. Increasingly IP is being proposed as the networking approach of the future — even in support of real time services, previously thought to be impossible or impractical. The advent of new protocols, such as the resource reservation protocol (RSVP), now makes it possible to attempt to reserve bandwidth through a routed network. Further work is needed to fully evaluate these claims under real networking conditions, and thereby understand the respective roles to be taken by ATM and IP in a future broadband network.

The broadband call control demonstrator platform could be used to investigate the control requirements of the underlying ATM network when supporting IP which itself is operating a resource reservation protocol session. The exact scope of this stage is for further study.

6.2 Open control

The connection manager functionality within the switch control has been implemented in such a way that other vendors’ ATM switches can be supported with only minimal change to the connection manager functionality. The next phase in this is to progress towards a truly open interface that will mean signalling and management software will be fully interchangeable between vendors’ ATM switches. This has the benefit of reducing the cost of this software development (through open competition) and reducing the time to upgrade software versions.

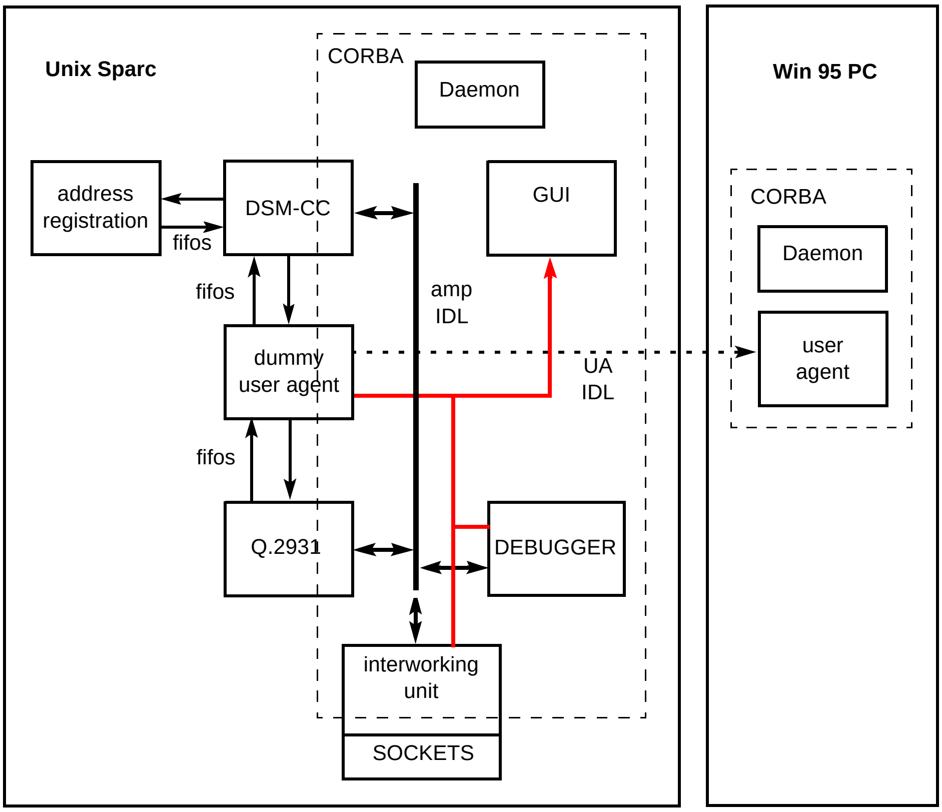

Fig 13 TINA retailer functionality.

6.3 *Network signalling*

The evolution of the broadband call control demonstrator platform has always been directed at making it as representative of a future BT broadband network as possible. One of the key steps in doing this is to support a multiple switch environment, with an implementation of network signalling. The demonstrator is currently based around a single ATM switch with only access signalling implemented.

Earlier work has provided an implementation of ITU-T B-ISUP network signalling which will be integrated with the current ITU-T Q.2931 access signalling via use of the ITU-T Q.2650 interworking Recommendation.

6.4 *Implementation of a TINA ConS interface*

The next phase in the exploration of interworking between TINA and DAVIC is to enhance the broadband call control demonstrator so that it supports one of the standard TINA reference points. The TINA ConS reference point exists between a connectivity provider (the network), and a TINA retailer. The TINA retailer can use this interface to instruct a network to establish a connection between a service provider and a service consumer. This would allow TINA-compliant service consumers and third party service suppliers to be supported by the ATM network of the demonstrator. Completion of such an interface would:

- verify the ConS interface,
- substantially enhance the functionality of the demonstrator,
- provide a tutorial platform for the understanding of TINA concepts,
- support non-TINA architectures that also work to a ConS interface.

Strictly the broadband call control demonstrator should support a ConS interface on the workstation controlling the ATM switch. However, it is likely to be more convenient to implement the functionality on the retailer as:

- this platform already had a CORBA platform and OO code,
- non-DAVIC-compliant developments are limited to a single component of the enterprise model,
- the retailer is likely to be a point of flexibility where DAVIC/TINA interworking issues could be addressed.

An overview of the proposed development is provided in Fig 14.

It is hoped this stage of the development will allow the demonstrator to be interworked to other demonstrators within BT which support the other side of a ConS interface.

6.5 *Collaborative development*

There are a number of future collaborative development work items under discussion or ongoing for 1997/98. This

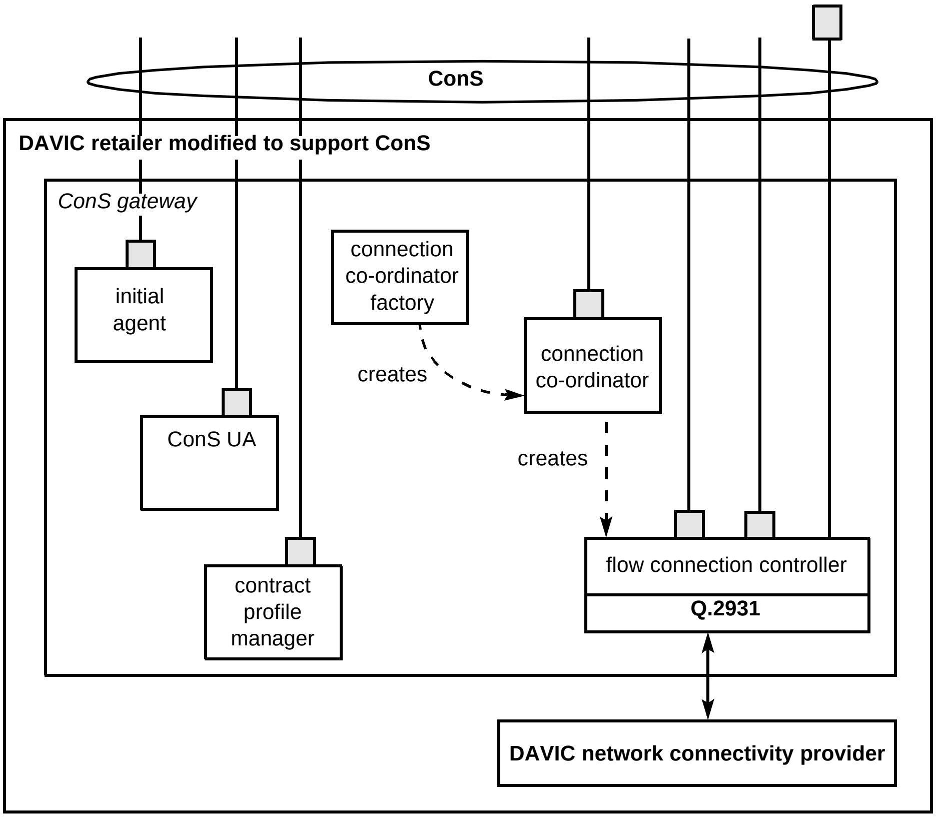

Fig 14 ConS interface development.

work involves collaboration with other areas of BT, other network operators and manufacturers of broadband equipment. Collaborative development of the demonstrator platform is discussed in more detail in King [6].

7. Conclusions

This paper has provided an overview of some of the work being undertaken at BT Laboratories to understand networking architectures and signalling systems. This is essential to provide a future broadband network capable of supporting on-demand ATM connections with guaranteed bandwidth and quality of service. It shows how the complexities of such a network can be better understood with the use of a demonstrator platform to model the information flows and processing requirements. By adopting a steady evolution of capabilities, it was possible to develop the platform, in a bottom-up approach, from simple connectivity into the support of complex services in a DAVIC-compliant manner.

In addition to providing a valuable learning tool for use within the broadband and data networks unit, the platform has been used as a means of promoting BT's expertise to its customers. The platform has been used to illustrate the future capabilities of BT's Cellstream service, where connections formed dynamically to a range of service providers will give increased choice to the customer and more efficient use of network resources. The demonstrator has also been positioned as a 'validation platform' for DAVIC/DVB architectures to allow closer working with vendors entering the consumer market with early prototypes of set-top boxes, etc.

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